a THE PROBLEM
Robustness up, abstention down
Q: "What flavor of jellybeans are in this package?"
label: unanswerable
Base → unanswerable
RKL → Cherry, Lime, …
GRPO → strawberry
Accuracy change vs. base (points)
+0.1
+6.9
+2.1
+7.6
base
-19.5
-15.7
Clean ▲
Corrupted ▲
Unanswerable ▼
b DIAGNOSIS
The model still knows
ℓ = 24
probe
Probe AUROC: is it unanswerable?
0.93
0.93
0.93
chance
Base
RKL
GRPO
Knows it can't answer
…but answers anyway
c OUR FIX
Restore abstention, keep robustness
+ λ·v
ℓ = 18
training-free
one vector, one layer
GRPO accuracy (%)
79.0
78.8
72.0
71.7
60.1
81.7
Clean
Corrupted
Unanswerable ▲
GRPO
+ steering
base
gain kept
Jellybeans → unanswerable